Personal Contribution Log — Emir Babayev

Project: Hardware, Networks & Behaviourist AI
Stage: Stage 1
Role: CEO / Project Coordinator and Hardware Lead

My contribution

- Helped establish and organise the shared GitHub project and its working structure.
- Prepared and maintained the project plan, task structure, learning sequence, and coordination documents.
- Developed the Hardware module with Ivan's research support. I assembled the explanations, examples, diagrams, questions, and answer material across the Hardware topics.
- Worked on the CPU and memory explanations, including the instruction trace and memory-hierarchy material, using beginner-friendly language.
- Used an LLM to help draft and simplify parts of the Hardware explanations and diagrams. I checked the technical meaning, corrected wording, and adapted the material for our learners instead of copying the output directly.
- Helped prepare the revised slides, video structure, and integration plan connecting Hardware, Networks, and Behaviourist AI.
- Took part in the team meeting on 25 October 2026 in the evening at B1. Together with Alimzhan, I identified that we had focused too much on topic content and had not made the project framework, continuation structure, and evidence clear enough.
- After that discussion, I helped reorganise the project structure, planning documents, review records, and Stage 1 presentation/video materials.

What I learned

Small Markdown files with a consistent structure are easier for a team to review and combine. Beginner explanations also require more work than technical notes because every assumption and unfamiliar term must be made visible. AI assistance can speed up drafting, but technical claims, examples, diagrams, and references still need human checking.

Remaining work

- Complete and record the final independent review of the revised Hardware material.
- Check the final four-minute classroom progress report and its offline USB package.
- Continue integrating the three modules into the Study Guide for Stage 2.

Evidence locations

- Hardware content: `content/Hardware/`
- Project plan and coordination: `project-management/PROJECT-PLAN.md` and `project-management/TASKS.md`
- Review record: `project-management/qa-log.md`
- Slides and classroom report plan: `slides/`
- Video plan and storyboards: `videos/`

Personal Learning and Work Log

Dobrydin Dmitrii | Student ID: 1820262226 | Team: team

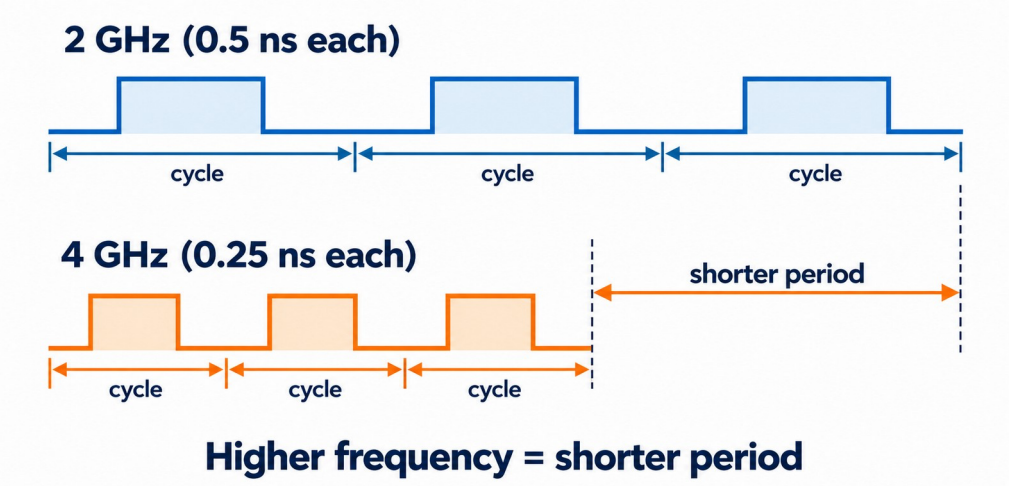
Period: 16–24 October 2026 | Retrospective update: 26 October 2026 | Beijing time

My main contribution was the Networks section. I did online background reading and prepared the SSH and DHCP material for our repository. I also created the original Hardware diagrams and helped with the team project plan.

23–24 October — Hardware diagrams

I created all the original Hardware diagrams for Stage 1, while Emir uploaded the images. The dates here refer to those uploads; the exact drawing dates were not recorded.

Evidence: My original log states “Created diagrams for Hardware” (65f8cc7). Emir’s image uploads are recorded in 40667d4 and 2830793.



An original Hardware figure: 2 GHz corresponds to 0.5 ns per cycle; 4 GHz to 0.25 ns. Source: the 24 October clock-period chapter.

24 October — SSH and DHCP

I selected the two topics, used the existing DNS/IP chapter format and uploaded the new files. The SSH chapter explains secure remote access, authentication and host keys. The DHCP chapter explains automatic configuration, DORA, leases and reservations.

Evidence: 20:19 — SSH, 9195c60: “The private key itself is not sent to the server.”
20:20 — DHCP, 4e40049: Discover → Offer → Request → Acknowledgment.

I also corrected SHH.md to SSH.md (7a4b9da). This reminded me to check filenames as carefully as the text. A useful distinction in the material is that DHCP supplies network settings, while DNS resolves names. Next, I want to add explained answers and keep better notes of the sources I read.

AI Use — record updated 26 October

Tool: OpenAI assistant. AI helped with the initial SSH/DHCP drafts and later editing of this log. I provided the topics and existing chapter examples. Request summary: two beginner texts using the DNS/IP structure. The uploaded excerpts above show the output; dates and excerpts were matched to Git for this update. Earlier checking notes were not retained. Emir’s shared Hardware log also records LLM help with diagrams.

Reference notes added 26 October: OpenSSH: ssh(1) • Microsoft Learn: DHCP overview • IETF: RFC 2131

PROJECT CONTRIBUTION RECORD
Personal Log
Emelyanov Ilya · GitHub: nsdmlk
PROJECT
Hardware, Networks & Behaviorist AI
EVIDENCE PERIOD
16–24 October 2026
REPOSITORY ROLE
CIO (coordination role)
This log summarizes work attributable to Ilya in the repository's Git history. Commit dates and changed files are used as evidence; the role listed in the README is context, not proof by itself.
Recorded contributions
16 Oct
Set up the repository
Created the initial project commit with the repository configuration, license, and README starting point. Evidence: a18f0d1.
17 Oct
Established early project structure and orientation
Expanded the README and added the initial Hardware and Networks introduction-file placeholders. Evidence: cec31da, 406b608, 0b99dd5.
17 Oct
Drafted the Behaviorist AI introduction
Added an introductory explanation of reinforcement learning and the initial architecture for this part of the learning material. Evidence: 9f17e84.
23 Oct
Developed core Behaviorist AI learning material
Added initial material on Q-learning, policies and rules, agent concepts, and Deep Q-Networks (DQN). These commits establish authorship of the drafts, not necessarily of the later revised versions. Evidence: f150636, efea261, 31febfe.
23 Oct
Updated project and robotics documentation
Made README changes and edited/reorganized the robotics-related files in the AI module. Evidence: 92f2d19, ebf3abd.
24 Oct
Corrected a Networks filename
Renamed the misspelled DHSCP file to DHCP. This was a filename correction; the commit does not establish authorship of the chapter's content. Also made a README update. Evidence: 1b96fff, 8c7b178.
Contribution summary
Repository evidence shows early project setup and documentation work, an initial Behaviorist AI introduction, draft content covering Q-learning, policies, agents and DQN, edits to robotics-related material, and a small Networks filename correction.
Attribution and limits. This is a repository-based record, not a timesheet. It does not estimate hours or claim unrecorded meetings, offline work, reviews, experiments, or final submission tasks. The large 26 October resubmission revision is not attributed to Ilya here; the project records identify it as team/AI-assisted work and the individual commits visible for that revision belong to other authors.
Prepared 26 October 2026 · Evidence: Git commit history and the repository README, project log, task list, and resubmission summary.

Personal Contribution Log

Nurpeissov Alimzhan | Student ID: 1820262236 | @Alimbeeeeeek

Stage 1 job log

26 October 2026 | Robotics chapter (ROBOT.md)

Completed work

• Explained how computer vision and reinforcement learning can support a robot's perception, decisions, and control, with practical examples.

• Described traditional robot algorithmization using predefined algorithms, sensor readings, conditional logic, and programmed actions, including FTC examples.

• Added five explanatory GIFs from a high-school FTC robotics project to illustrate the robot system and its behaviour.

Personal Learning and Work Log

Shamshiev Adil / 车一凡 | Student ID: 1820262243 | Team: team

Period: through 24 October 2026 | Role: video lead | GitHub: protoadil | Beijing time

My main contribution was the Networks section of our repository.

23 October — Networks chapters

I committed six basic Networks chapters: DNS, HTTP and security, LAN/WAN/PAN, the OSI model, routing and IP, and TCP and UDP. The introduction file was an existing draft by a teammate, so I do not claim it here.

Evidence: commit 4f7ae86 “network” (DNS.md, HTTP-AND-SECURITY.md, LAN-WAN-PAN.md, OSI-MODEL.md, ROUTING-AND-IP.md, TCP-AND-UDP.md).

24 October — Networks introduction

I wrote a basic introduction to Networks so a beginner knows what the section covers.

Evidence: commit 020eabb “Stage 1: wrote basic introduction to networks”.

AI Use

I used Claude Code and DeepSeek to help me understand and explain Networks topics, and I also searched online for background reading.

What I learned

DNS. Computers connect using IP addresses, but people use names. DNS translates a name into an IP address: a resolver asks other servers in turn until it reaches the one that knows the answer, and keeps the result for a while so the next lookup is faster.

TCP and UDP. TCP sets up a connection first, resends lost data and keeps it in order, so it suits web pages and files. UDP skips all of that, which makes it faster but less reliable, so it suits things like video calls where a late packet is useless anyway.

OSI model. It splits networking into seven layers. It is useful less as something to memorise and more as a way to find where a problem is, for example a wrong IP address (network layer) versus a website that will not load (application layer).

Personal Contribution Log — Ivan Maidachenko

GitHub: @ivanelloj | Team: team (BIT) | Project: Hardware, Networks & Behaviourist AI | Stage 1
Role: Hardware Research and Visual Support (README: Visual design lead) | Evidence period: 16–26 October 2026 | Prepared 26 October 2026, Beijing time

Summary

My main contribution was background research for the Hardware module. I collected and checked source material for the Hardware topics and passed it to Emir, who assembled the module and uploaded the files. I attended every team meeting, including the 25 October architecture review at B1. In line with the project plan, I record here that the author of the research and the uploader of the files are different people: my research is contained in **content/Hardware/**, but the commits there are under Emir's name, so this log does not claim individual commits.

Recorded contributions

Date	Activity	Evidence / location
16–22 Oct	Joined the repository and team planning; agreed with Emir on the Hardware section list (CPU, fetch–decode–execute, memory hierarchy, how a program runs, performance, pipelining) and started reading sources for each.	Team roles in README.md (8602c80, 18 Oct); Hardware intro draft 9b6f812 (22 Oct) based on the agreed list
23 Oct	Supplied research notes used for the first Hardware upload: five parts of a computer, PC/IR, registers and ALU, control unit, buses, fetch–decode–execute, memory hierarchy, RAM and storage. Checked that terms were defined before use.	Section folders 01–04 under content/Hardware/; uploads by Emir, e.g. 18afca2, 4beee1c, e825ec9, 8c5784c, 6f38f29 (23 Oct, 11:33–12:01)
23–24 Oct	Researched the performance and pipelining topics: GHz vs. speed, instruction count and CPI, clock period, worked CPU-time calculation, assembly-line analogy and throughput. Reviewed the GPU/accelerator and NIC latency/bandwidth pages for visual clarity.	Section folders 06–08; uploads 07aa939, 3daa991, 87ae532, d49beaf (23 Oct), 819961d, 584975f, 9979bb8 (24 Oct)
25 Oct (evening, B1)	Attended the full-team review of the first submission (19/30). Agreed with the finding that we had focused on topic content and not on the project framework and evidence; took the action to support visual consistency of the revised Hardware diagrams.	project-management/peer-review-record.md; TASKS.md A3 and A6 ("Emir with Ivan")
26 Oct	Checked the revised Hardware material and the new local SVG figures (h01–h09) against the research notes; confirmed the figure list in the asset provenance record. Wrote this log.	content/Hardware/assets/, docs/ASSET-PROVENANCE.md, revision commit 2b9c960

What I learned

A computer has five parts: CPU, memory, storage, input and output; a NIC is an I/O device. Inside the CPU the PC holds the address of the next instruction and the IR holds the one being executed; the control unit sequences the fetch–decode–execute cycle, the ALU does the arithmetic, and registers are the fastest storage. Memory is layered (registers, cache, RAM, SSD/HDD) because fast memory is small and expensive, so a cache miss costs far more than a hit. A program is just instructions stored in memory: a compiler turns $a = b + c$ into LOAD, ADD and STORE. GHz alone is not speed — CPU time = instruction count × CPI × clock period, so 2 GHz means 0.5 ns per cycle and a lower CPI can beat a higher clock. Pipelining overlaps instructions like an assembly line: it raises throughput, not the latency of one instruction. GPUs help AI because many identical operations run in parallel, while network latency and bandwidth limit how fast data reaches them.

AI use

An AI assistant was used during the research to find and compare candidate sources for each Hardware topic; I then read the sources myself and chose which explanations and figures to pass on, checking the technical claims before handing them to Emir. No AI was used to write this log. Emir's log separately records LLM help in drafting and simplifying the Hardware explanations built from the research.

Remaining work

- Support Alimzhan's independent content check of one Hardware activity (task A11) by supplying the sources behind the ADD/LOAD trace and CPI example.
- Prepare readable diagram captures and overlays for the four-minute classroom report video (video-plan: visual design role).
- Commit Stage 2 Hardware research and figure revisions directly to the repository so authorship is recorded in Git.

Attribution and limits: this is a repository-and-meeting-based record, not a timesheet. It does not estimate hours. Evidence: Git history of github.com/nsdmlk/bit-ai-and-cs, README.md, PROJECT-PLAN.md, TASKS.md, peer-review-record.md, ASSET-PROVENANCE.md.